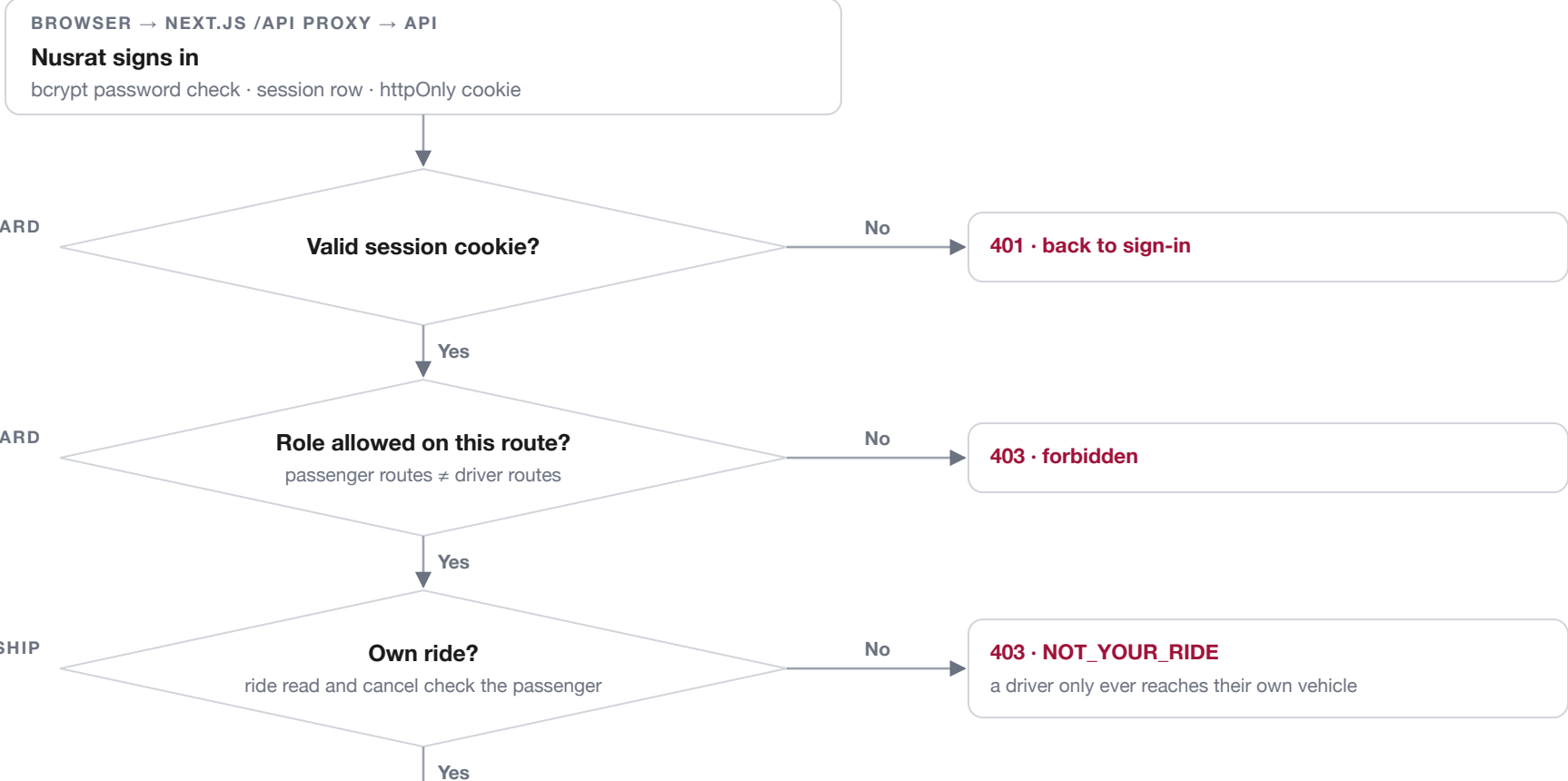


Dhaka Tesla Pool · Ride Flow

How one ride moves through the system · a Tesla picks people up along its route · side notes cover concurrent and edge cases

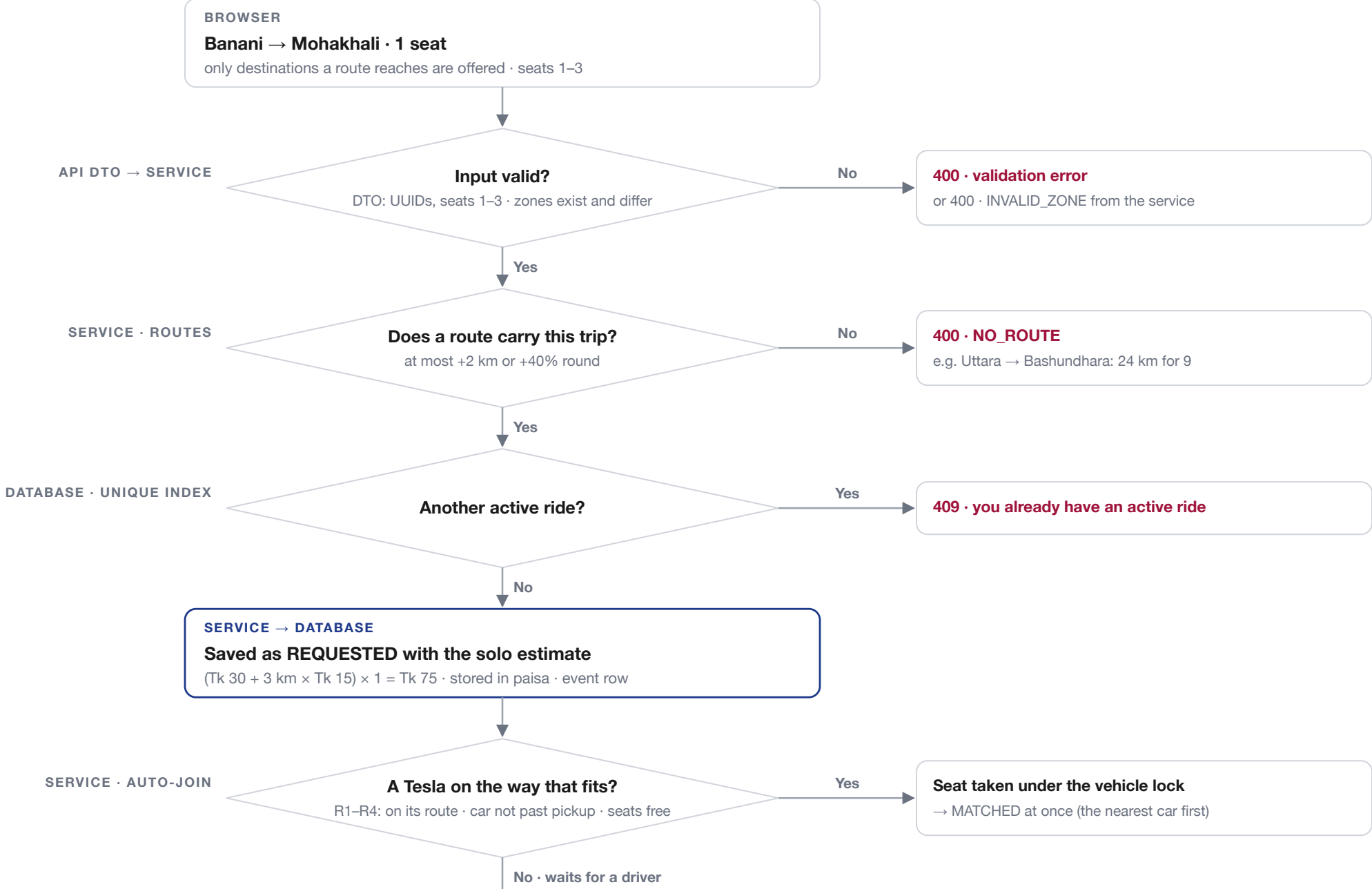
1 SIGN IN



Sign up

Passengers only; drivers are seeded
Unique email (stored lower case)
Password hashed with bcrypt
Sign-up and login: 5 per minute

2 REQUEST A RIDE



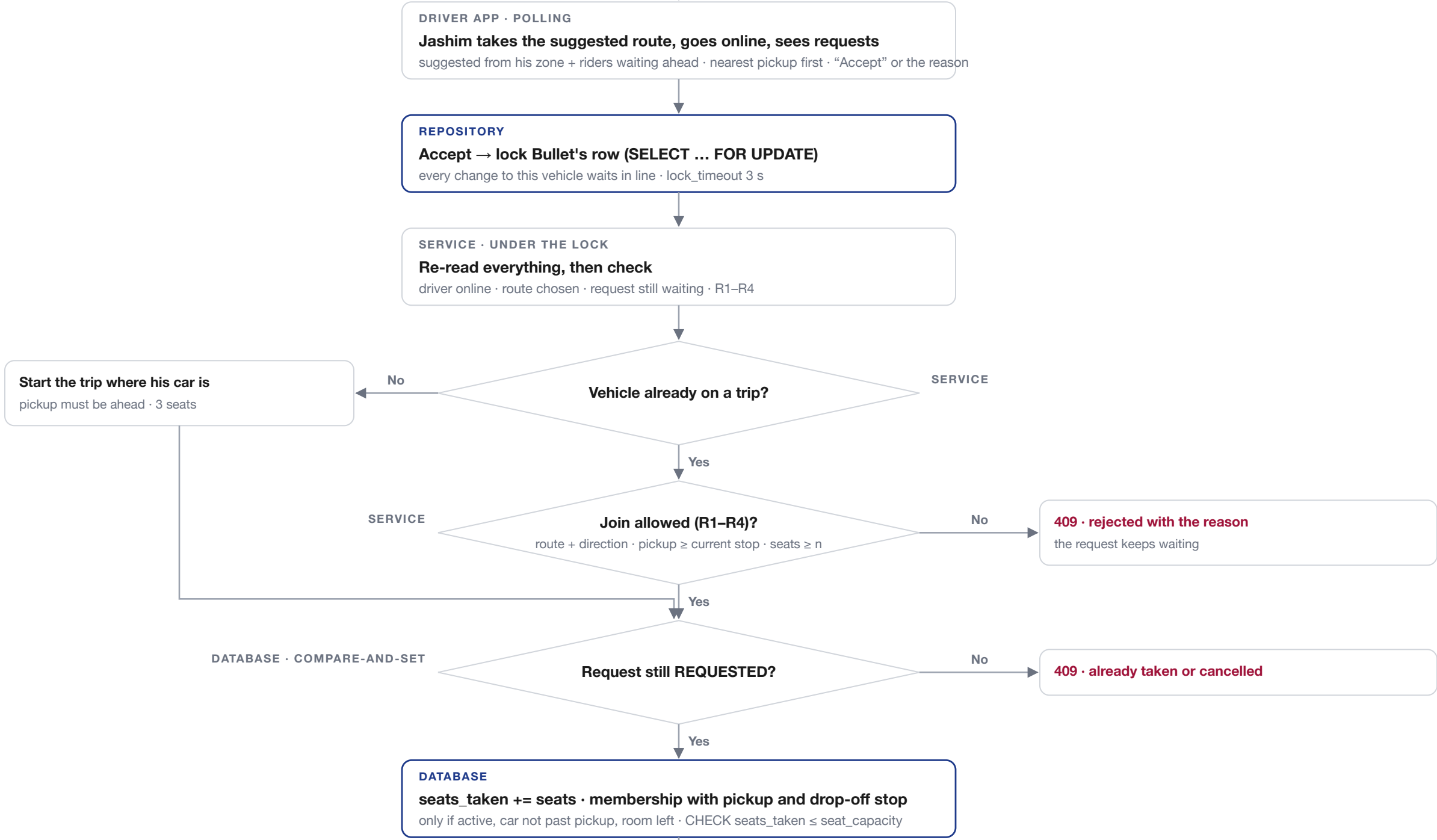
Routes (R1)

6 routes: 3 lines, both directions
At most +2 km or +40% round
Rafiq: 6 km for 4 km ✓
Uttara → Bashundhara: 24 for 9 ✗

Who pays, who earns

Rider: (Tk 30 + km × Tk 15) × seats
–20% if shared a hop · locked at drop-off
Driver: Tk 10/km carried + Tk 20/pickup
Platform keeps the rest (≥ Tk 10)

3 MATCHING



Last seat, many riders at once

Nusrat holds 2 of Bullet's 3 seats
20 riders ask for the last one
All wait for Bullet's lock
Exactly 1 wins · 19 keep waiting

No trip yet, two accepts at once

First accept starts the trip
Second waits, then joins it
One trip, two passengers

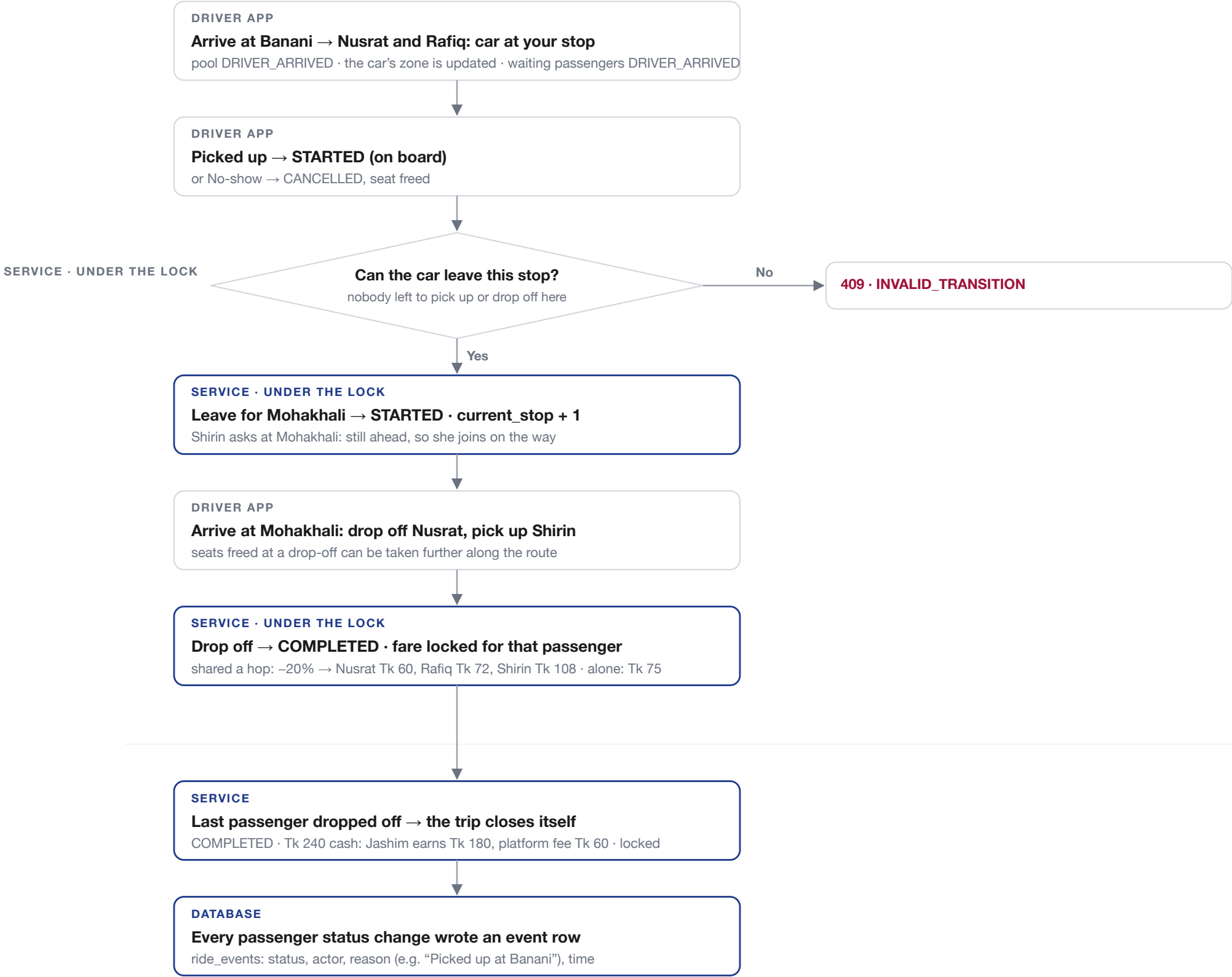
Retries and timeouts

Lock wait > 3 s → 503 BUSY
Auto-join is best effort: if busy,
the ride just keeps waiting

Database guarantees

seats_taken ≤ seat_capacity
pickup_stop < droptoff_stop
1 active pool per vehicle
1 active membership per ride
1 active ride per passenger

4 THE TRIP, STOP BY STOP



Route, location, online

System suggests, driver confirms
Location follows the stops
Route changes only between trips
Offline refused while a pool is active

Leave vs join at the same stop

Car at Mohakhali, Shirin asks there
Both wait for Bullet's lock
Join first → the car waits for her
Leave first → stop is behind: she waits

What the passenger sees

Polls her ride every 3 s (history 10 s)
The route with the car on it
Own status and own fare only
Co-riders' first names only

Cancellation rules

Passenger: until picked up
→ seats freed at once
Driver at the stop: no-show
Driver: before the first pickup →
passengers back to REQUESTED
Last one out → the trip closes